

RESEARCH ARTICLE

Semantic Identifiability under Band-Order Ambiguity in PlanetScope Landslide Segmentation

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ABSTRACT

Released HR-GLDD NumPy arrays lack authoritative visible-band metadata: direct notebook display supports Red-Green-Blue-NIR (RGBN), whereas native four-band PlanetScope supports Blue-Green-Red-NIR (BGRN). We apply a finite identified-set certificate: for a signed mechanism contrast, its envelope spans all metadata-consistent schemas, and direction is retained only when that envelope excludes zero. Complete RGBN/BGRN runs give index-input-versus-raw F1 envelope $[-0.21, -0.01]$ points and equal-mass index-versus-plain F1 envelope $[-0.41, -0.33]$, but the boundary-F1 envelope $[-0.05, +0.38]$ crosses zero. Generic boundary weighting remains positive, $[+0.96, +1.13]$, with all 18 individual effects positive. A post-hoc CAS analysis fails three of four criteria. An internally precommitted LRD run also fails its frozen criteria, but weak validation, trigger-family overlap, and empty-crop sensitivity make it failed/indeterminate rather than confirmation. HR-GLDD results remain descriptive because event and coordinate identities are absent.

KEYWORDS

landslide segmentation; PlanetScope; band-order ambiguity; schema robustness; boundary weighting; mechanism attribution

Word count: 2,788 (Pandoc; abstract-conclusion, excluding tables, figures, displayed equations, references, declarations).

1. Introduction

High-resolution optical imagery enables detailed landslide mapping, but roads, river deposits, clearings, and exposed rock share similar structure. HR-GLDD provides 3 m PlanetScope patches (Meena et al. 2023); CAS provides multisensor RGB imagery (Xu et al. 2024); Landslide Reference Data (LRD) supplies named Sentinel-2 events (Orynbaikyzy et al. 2025).

NDVI is attractive because fresh failures can disrupt vegetation, but roads, agriculture, riverbanks, and shadows also create index edges. An index-modulated boundary loss changes both boundary emphasis and spectral modulation unless ordinary boundary weighting is controlled. Array semantics are equally load-bearing: assigning Red to the wrong column changes both NDVI and raw-edge controls.

This distinction matters for remote-sensing image processing. Devara et al. compared an empirical extraction method with U-Net for Sentinel-2 landslides (Devara,

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Maurya, and Dwivedi 2024), while Wang et al. designed edge enhancement for remote-sensing semantic segmentation (Wang, Li, and Shi 2025). These studies motivate boundary-specific outcomes and make generic edge and boundary controls necessary before a vegetation-specific interpretation can be claimed.

Known-order channel permutations test adaptability (Li et al. 2025), not unresolved release metadata. We apply established identified-set logic to a full HR-GLDD physical-schema audit with controlled reruns; we do not claim a new sensitivity primitive.

We therefore evaluate components and semantic assumptions rather than presuming a mechanism. Our contributions are:

- (1) a semantic-identifiability certificate that reports whether signed contrasts retain direction over every metadata-consistent schema;
- (2) an executable audit showing that HR-GLDD columns 0 and 2 remain ambiguously Red/Blue, followed by complete RGBN and BGRN sensitivity runs;
- (3) capacity- and weight-matched zero, raw-edge, candidate-index, plain-boundary, and index-modulated controls;
- (4) outcomes selected without test-label threshold tuning, plus post-hoc CAS and internally precommitted LRD stress analyses.

The certificate identifies negative index-versus-generic F1 directions and a positive generic boundary direction, but leaves the index-modulated boundary direction unidentified. External analyses do not confirm transfer, and LRD defects make its result indeterminate rather than a robust rejection.

2. Related work

2.1. *Landslide segmentation*

U-Net and residual-block U-Net architectures provide standard segmentation baselines (Ronneberger, Fischer, and Brox 2015, Zhang, Liu, and Wang 2018). HR-GLDD is a PlanetScope benchmark (Meena et al. 2023), while CAS provides multisensor RGB patches (Xu et al. 2024). In Remote Sensing Letters, Devara et al. compare U-Net with an empirical Sentinel-2 rule (Devara, Maurya, and Dwivedi 2024). These performance studies do not isolate the active component.

2.2. *Spectral and edge guidance*

NDVI is defined from Red and NIR reflectance and can reduce a shared multiplicative illumination factor under an idealized model (Matsushita et al. 2007), while TriGEFNet reports an NDVI-guided benefit (Zhang et al. 2026). Spectral indices have also been combined directly with edge extraction (Zhang et al. 2023). SAEENet and weak-boundary extraction add generic boundary guidance (Wang, Li, and Shi 2025, Wang et al. 2026). Li et al. deliberately permute six known HLS bands to test foundation-model adaptability (Li et al. 2025); we instead retain unresolved released-array identity and test attribution with zero, raw, and equal-mass controls.

2.3. Finite sensitivity and identified sets

Multiverse analysis propagates reasonable coding choices (Steege et al. 2016), while partial identification retains sets under structural uncertainty (Manski, Sanstad, and DeCanio 2021). Qin and Tong make family-relative identified sets executable for evaluation claims (Qin and Tong 2026). Our min-max sign rule applies that established primitive; the remote-sensing delta is the physical-schema construction, full retraining, same-class controls, and hash-bound HR-GLDD claim record.

2.4. Boundary weighting and evaluation

Pixel-wise boundary emphasis predates this application in U-Net (Ronneberger, Fischer, and Brox 2015). LSL-SS-Net uses a landslide-specific level-set loss, while E-MSLDNet reports edge-aware CAS segmentation with boundary metrics (Yang et al. 2024, Zhao et al. 2025). We do not propose another boundary architecture or loss; we test whether index-specific attribution survives plain-boundary same-class control and unresolved physical schema.

3. Controlled methods

3.1. Band semantics and spectral controls

Let $\mathbf{X} \in \mathbb{R}^{B \times 4 \times H \times W}$ denote the released HR-GLDD arrays. Green and NIR occupy columns 1 and 3 under both plausible orders; Red is column 0 under RGBN and column 2 under native-product BGRN. The paper names the selected bands but not their NumPy positions, while the official notebook directly displays columns 0–2 as RGB without exposing array construction. We therefore run the full matrix for each candidate order $o \in \{\text{RGBN}, \text{BGRN}\}$ and define the candidate normalized-difference index

$$V_o = \frac{X_{\text{NIR}} - X_{\text{Red}(o)}}{X_{\text{NIR}} + X_{\text{Red}(o)} + 10^{-6}}. \quad (1)$$

We compare three two-channel controls:

$$C_{\text{zero}} = (0, 0), \quad (2)$$

$$C_{\text{raw},o} = (|\nabla X_{\text{Red}(o)}|, |\nabla X_{\text{NIR}}|), \quad (3)$$

$$C_{\text{index},o} = (V_o, |\nabla V_o|). \quad (4)$$

The zero control preserves attention parameters while removing external spectral information. The raw control tests whether generic optical edges explain any candidate-index advantage. Every recorded HR-GLDD output declares its candidate order; neither candidate is declared authoritative.

3.2. Semantic-identifiability certificate

Let $\mathcal{O}$ be the finite set of metadata-consistent schemas and $\delta(o)$ a signed contrast under schema o . We define its semantic envelope

$$\mathcal{E}_\delta = \left[\min_{o \in \mathcal{O}} \delta(o), \max_{o \in \mathcal{O}} \delta(o) \right]. \quad (5)$$

The effect direction is semantically identified only when $0 \notin \mathcal{E}_\delta$; otherwise it is direction-unidentified. Here $\mathcal{O} = \{\text{RGBN}, \text{BGRN}\}$. This is a finite schema sensitivity certificate over descriptive means, not a confidence interval or a claim that no unlisted schema exists.

3.3. Capacity-matched residual attention

All controlled-attention models use the same residual encoder-decoder with channels (32, 64, 128, 256) and three skip gates. At scale l , the two control maps are bilinearly resized and concatenated with encoder features F_l :

$$M_l = \sigma[\text{Conv}_{1 \times 1}(\text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}[F_l, C_l])))], \quad (6)$$

$$F_{l,\text{out}} = F_l \odot M_l + F_l. \quad (7)$$

Every control mode therefore has 2,114,084 trainable parameters. U-Net and ResU-Net are included as architectural baselines.

3.4. Loss controls

For pixel i , the unreduced base-loss map combines binary cross-entropy and focal loss (Lin et al. 2017), with focal positive-class weight $\alpha = 0.35$ and focusing exponent 2:

$$\ell_{\text{base},i} = \frac{1}{2}\ell_{\text{BCE},i} + \frac{1}{2}\ell_{\text{focal},i}. \quad (8)$$

Let $B_{\text{GT}} = \text{Dilate}_{3 \times 3}(Y) - \text{Erode}_{3 \times 3}(Y)$ denote the two-sided, one-pixel-radius Chebyshev morphological boundary band. We compare:

$$L_{\text{plain}} = \text{mean}_i\{\ell_{\text{base},i}[1 + \gamma B_{\text{GT},i}]\}, \quad (9)$$

$$L_{\text{index},o} = \text{mean}_i\{\ell_{\text{base},i}[1 + \gamma \widetilde{B_{\text{GT}}|\nabla V_o|_i}]\}, \quad (10)$$

where $\gamma = 2$. For an informative index-boundary signal, the tilde rescales it within each image to the mean mass of B_{GT} ; when its mean is at most 10^{-6} , it falls back to B_{GT} . Under both candidate orders, 8 of 1,119 training images use this fallback and the maximum mass mismatch is 2.98×10^{-8} . Thus L_{plain} versus $L_{\text{index},o}$ changes spatial allocation without changing average added weight.

3.5. Evaluation quantities

We compute pooled F1, intersection over union, precision, recall, background false-positive rate (FPR), and per-tile macro F1; the compact main table reports the quantities used by the claims. Boundary F1 is computed per tile after extracting predicted and reference boundary bands; a boundary pixel matches when the opposite boundary lies within a one-pixel Chebyshev (8-neighbour) tolerance.

For an operating-point comparison, U-Net recall at threshold 0.5 is measured on validation data. Every arm, including U-Net, is then mapped to the exact validation-score breakpoint nearest that recall, with ties assigned to the higher threshold, and the selected threshold is applied unchanged to test probabilities. Test labels do not select thresholds.

To isolate spatial degradation without changing sensor or region, each PlanetScope test patch is antialiased from 128×128 to 38×38 (approximately the 3 m-to-10 m ratio) and restored to 128×128 before inference. This experiment measures controlled information loss; it is not a simulation of Sentinel-2 radiometry.

4. Experimental design

4.1. Dataset and scope of inference

We use the released HR-GLDD arrays (Meena et al. 2023): 1,119 training, 284 validation, and 355 test patches of 128×128 pixels at 3 m ground sampling distance. Landslide pixels comprise 10.09%, 10.17%, and 10.92% of the three splits. The release contains imagery and masks but no event identifier, source-scene identifier, or coordinate for each array row. Separate patches therefore cannot be demonstrated to be independent. Results are descriptive across three training seeds; we do not calculate tile-bootstrap confidence intervals, p-values, spatial-block statistics, or event-held-out performance.

The official test set had also been inspected during earlier method development. The present experiment is consequently treated as corrective evaluation rather than untouched confirmation. It can compare controlled configurations on the released benchmark but cannot establish population-level geographic generalization.

4.2. Factorial controls

The residual-attention surface crosses zero, raw-edge, and candidate-index controls with base and plain boundary-weighted losses; one additional index arm uses equal-mass index-modulated weighting. U-Net and ResU-Net bring the total to nine configurations. We train all nine with three seeds under both RGBN and BGRN candidates (54 runs).

All arms use AdamW (Loshchilov and Hutter 2019) with learning rate 1.5×10^{-3} , weight decay 10^{-4} , cosine decay to 10^{-5} , batch size 32, and 15 epochs. Validation F1 selects the epoch. Seeds 42, 43, and 44 use identical schedules and deterministic data-loader seeds. Probabilities are stored as float32 arrays; an independent script recomputes the reported default-threshold metrics without importing the training or reporting functions.

4.3. *Post-hoc CAS sensitivity*

We compare base and plain-boundary loss for the same RGB zero-control model on designated CAS regions (Xu et al. 2024). The protocol was corrected after execution, so this is post-hoc sensitivity without confirmation credit; Supplementary Method S1 gives sampling and deviations.

4.4. *Internally precommitted LRD stress test*

LRD supplies named Sentinel-2 bands and 28 EIDs (Orynbaikyzy et al. 2025). A local 16/6/6 protocol preceded validation/protected access, but no third-party timestamp exists and PH0003 shares a trigger family with protected PH0001/PH0004. Supplementary Method S2 gives the frozen design and limitations.

4.5. *Interpretation rules*

Input specificity requires the candidate index to outperform zero and raw controls on the same endpoint under both orders. Spectral loss specificity requires index modulation to beat equal-mass plain weighting and retain effect direction across orders; boundary F1 is separate. The certificate covers index-versus-raw F1, index-versus-plain F1 and boundary F1, and plain-boundary-versus-base F1. Complete-system improvement overrides neither control. Resolution attribution is restricted to same-sensor degradation. The LRD run is confirmation-eligible only if timing, trigger independence, validation performance, and endpoint sensitivity are defensible; they are not.

5. Results

5.1. *Band-order sensitivity and matched controls*

Table 1. Three-seed mean performance under both plausible HR-GLDD orders. BF1 is boundary F1 with one-pixel Chebyshev tolerance. Neither order is declared authoritative.

Configuration	RGBN F1	BGRN F1	RGBN BF1	BGRN BF1
U-Net/base	66.92	66.92	49.62	49.62
ResU-Net/base	68.55	68.55	52.93	52.93
Attn-zero/base	68.88	68.88	53.17	53.17
Attn-raw/base	69.05	69.13	52.92	53.15
Attn-index/base	69.04	68.92	52.76	52.83
Attn-zero/plain-boundary	69.99	69.99	56.04	56.04
Attn-raw/plain-boundary	69.88	70.19	55.85	56.15
Attn-index/plain-boundary	69.98	70.12	55.96	56.20
Attn-index/index-boundary	69.58	69.78	56.34	56.15

Table 1 reports every configuration under both candidate orders; Supplementary Table S3 gives detailed RGBN-candidate operating metrics. With base loss, candidate-index attention reaches 69.04% F1 under RGBN and 68.92% under BGRN. Raw candidate-Red/NIR edges reach 69.05% and 69.13%, while the capacity-matched zero control reaches 68.88% under both. Candidate-index input specificity is therefore unsupported regardless of order. All contrasts and envelopes below use unrounded run-level values; subtracting the table’s two-decimal cells can differ by 0.01 point.

Plain boundary weighting raises F1 in every seed and control mode under both orders. Mean F1 spans 69.88–70.19% and boundary F1 55.85–56.20%. Equal-mass index modulation changes F1 by -0.41 points under RGBN and -0.33 under BGRN. Its boundary-F1 contrast changes from $+0.38$ to -0.05 points, so spectral modulation is not order-robust. The order-stable result is generic boundary emphasis, not spectral identity.

5.2. *Semantic-identifiability certificate*

The index-input-versus-raw F1 envelope is $[-0.21, -0.01]$ points and the index-versus-plain-loss F1 envelope $[-0.41, -0.33]$, identifying negative directions over the two admissible schemas. The corresponding boundary-F1 envelope $[-0.05, +0.38]$ crosses zero and is direction-unidentified. Generic plain-boundary F1 has envelope $[+0.96, +1.13]$; all 18 individual effects lie between $+0.42$ and $+1.62$ points. Supplementary Table S4 is the generated certificate.

5.3. *External analyses*

The post-hoc CAS sensitivity analysis fails three of four quantitative criteria. Across three designated regions and seeds, plain boundary weighting changes mean region-macro F1 by -2.19 points and boundary F1 by -1.86 points; seed F1 effects are -10.42 , $+1.22$, and $+2.62$ points. Supplementary Table S1 reports every contrast. Because the protocol was first committed after execution, this result receives no confirmation credit.

The internally precommitted LRD run fails all four frozen criteria, but is not confirmation-grade. Maximum validation F1 is only 0.008–0.151; protected event-macro F1 spans 0.007–0.283 for base and 0.009–0.132 for boundary weighting. Under selected thresholds, mean F1 changes by -4.73 points. The frozen per-crop boundary convention yields -24.10 points, but crop-macro empty/empty scoring of zero yields -0.15 , pooled selected-threshold scoring -4.92 , and pooled common-0.5 scoring -0.55 . Development PH0003 and protected PH0001/PH0004 also share a trigger family, leaving four non-overlapping protected EIDs. This external evidence is failed/indeterminate, not a robust transfer rejection. Supplementary Table S2 and Method S2 report the frozen and sensitivity results.

5.4. *Operating point and spatial degradation*

At threshold 0.5, boundary-weighted residual-attention FPR spans 2.47–2.50% across orders. Validation-selected nearest recall breakpoints match exactly in 53 of 54 order-specific runs; the RGBN index/plain-boundary seed-42 run differs by 2.11×10^{-6} . Combined matched FPR and precision ranges across the seven residual-attention configurations are 1.97–2.01% and 78.41–78.74%. These differences do not support index-specific false-positive suppression.

Controlled $128 \rightarrow 38 \rightarrow 128$ degradation lowers F1 by 4.33–5.20 points across both orders. Raw-edge plain-boundary F1 changes from 69.88 to 65.37% under RGBN and from 70.19 to 65.56% under BGRN. This identifies same-sensor spatial-information sensitivity, not cross-sensor causation.

6. Discussion

6.1. *What the controls change*

The semantic audit changes the evidentiary question. HR-GLDD’s released NumPy order is not authoritative, so an NDVI interpretation cannot rest on one assumed Red column. The certificate separates schema-invariant directions from a schema-contingent one: index-versus-generic F1 contrasts are negative throughout their envelopes, whereas the index-modulated boundary envelope crosses zero. Spectral boundary direction is therefore not identified by the released semantics.

Generic boundary weighting is more stable within HR-GLDD: all 18 seed/control/order comparisons improve F1. Neither external analysis confirms transfer. CAS is post hoc with mixed seed signs; LRD fails its frozen criteria but has weak validation, empty-crop endpoint instability, and trigger-family leakage. These analyses make transfer unsupported, not robustly refuted.

This does not imply that NDVI is universally unhelpful. Devara et al. report useful spectral processing (Devara, Maurya, and Dwivedi 2024), and TriGEFNet reports an NDVI-guided multimodal benefit (Zhang et al. 2026). Our contribution is diagnostic: explicit semantic uncertainty and same-class controls, not competitive performance or confirmed transfer.

6.2. *Interpretation under incomplete observability*

Band-ratio cancellation is exact only for a shared multiplicative irradiance term. Mountain scenes can additionally contain diffuse illumination, cast shadow, adjacency, bidirectional reflectance, and atmospheric effects. An NDVI gradient also responds to any vegetation discontinuity. The experiment contains no illumination labels or background-type annotations; it cannot validate topographic invariance or isolate scarps from roads and riverbanks.

More generally, methodological claims in incompletely observed environmental systems require explicit assumptions and dataset-specific calibration (Varotsos and Zhou 2026). We apply that principle by separating same-sensor spatial degradation from multi-factor cross-dataset shift and withholding event-level inference where group metadata are absent.

6.3. *Implications for benchmark practice*

Four practical recommendations follow. First, multispectral array releases should carry machine-readable channel names rather than relying on sensor conventions. Second, claims about geographic transfer or confidence intervals should be tied to event, scene, or spatial-block identifiers; a patch index is not an independence unit. Third, mechanism studies should include generic same-class controls. For spectral attention this means zero and raw-edge controls; for a specialized boundary loss it means ordinary boundary weighting with matched total mass. Fourth, unresolved schemas should be propagated into a machine-readable semantic envelope rather than collapsed to one convenient interpretation.

The strongest HR-GLDD configuration is compact, but the external sensitivity failure prevents presenting it as a generally useful boundary-aware model. Component latency and parameter count characterize implementation cost only; they are not evidence of an end-to-end emergency workflow.

7. Limitations

HR-GLDD does not authoritatively bind Red and Blue to columns 0 and 2. The certificate is finite over these two candidate schemas and descriptive means; it neither identifies wavelengths nor supplies statistical uncertainty. The arrays also omit event, scene, and coordinate mappings, precluding event-clustered or spatial inference; prior test inspection makes the evaluation corrective.

Binary masks omit background categories, so Supplementary Figure S1 supports visual inspection but not category-specific error rates. The 10-m control models PlanetScope smoothing and resampling, not Sentinel-2 radiometry or geography.

The post-hoc CAS analysis jointly changes sensor, resolution, geography, and annotation. LRD has no independently timestamped pre-access record; two protected EIDs share a trigger family with development; 84.1% of protected crops are empty; and validation F1 is weak. Its frozen criteria fail, but the external conclusion is indeterminate.

Three seeds quantify optimization variation, not independent events. No regional mosaic, acquisition pipeline, geospatial reconstruction, hardware payload, or responder workflow was evaluated; operational deployment and broad geographic generalization remain outside scope.

8. Conclusion

HR-GLDD's visible-band order is not authoritatively documented. Its semantic-identifiability certificate assigns negative envelopes to index-versus-generic F1 contrasts, a positive envelope to generic boundary F1, and an envelope crossing zero to index-modulated boundary F1. No spectral-specific boundary direction is identified.

Ordinary boundary weighting improves every HR-GLDD seed/control/order comparison. Neither external analysis confirms transfer: CAS is post hoc, while the internally precommitted LRD run is indeterminate because validation, endpoint, and trigger-family checks fail. Generic boundary weighting is therefore dataset-local evidence, not a transferable prior.

Machine-readable semantics, schema-propagated certificates, same-class controls, independently timestamped protocols, and true independence units must precede physical or generalization interpretations.

Acknowledgements. Cursor assisted software development, reference checking, and language refinement. The authors independently verified all source, code, references, and claims and take full responsibility.

Funding. No specific funding was received for this work.

Disclosure statement. No potential conflict of interest was reported by the authors.

Data availability statement

The HR-GLDD arrays are at <https://doi.org/10.5281/zenodo.7189381>; LRD is at <https://doi.org/10.5281/zenodo.17007637>. Code, summaries, output hashes,

and the complete source-tarball/history assets are in <https://github.com/akssha74/S3Net-Landslide> (release v5.3.0). CAS imagery remains at <https://doi.org/10.5281/zenodo.10294997>; source terms bind.

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